GRaD-Nav usability decision

NO-GO for calibrated R4R camera replay with this released gate_mid package

Native rendering and dynamics work as research components.
Capture/checkpoint correspondence remains unresolved.

One controlled version comparison. Original weights unchanged.

Inputs, outputs and E2E scope

Scene config + Gaussian checkpoint + camera pose/K → RGB, depth, alpha

Body rates + normalized thrust + state → next simulated state

PD controller → dynamics → achieved camera → native images/video

Original learned visual policy → actions/outcomes: blocked by weights

The demonstrated controller consumes height/velocity, not rendered images. Calibrated replay, visual autonomy and outdoor fidelity are separate gates.

Checkpoint restoration and camera corrections

Strict loading: no missing keys and no unexpected keys in either version

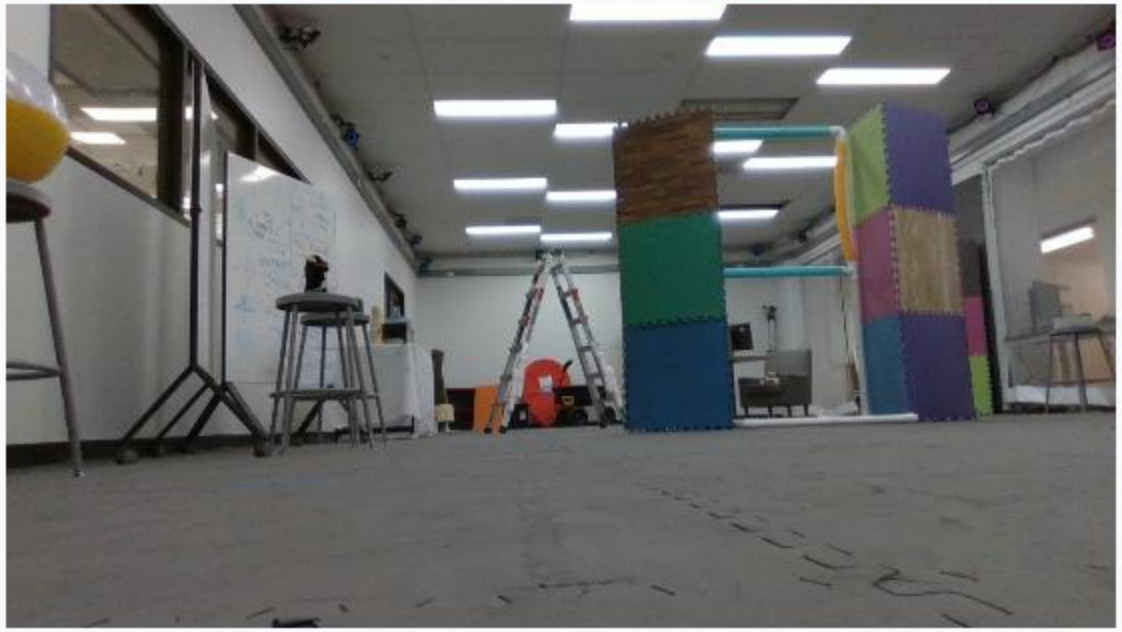
All six Gaussian arrays and all saved model tensors restored exactly

270 saved pose corrections; median translation 0.01597 scene units

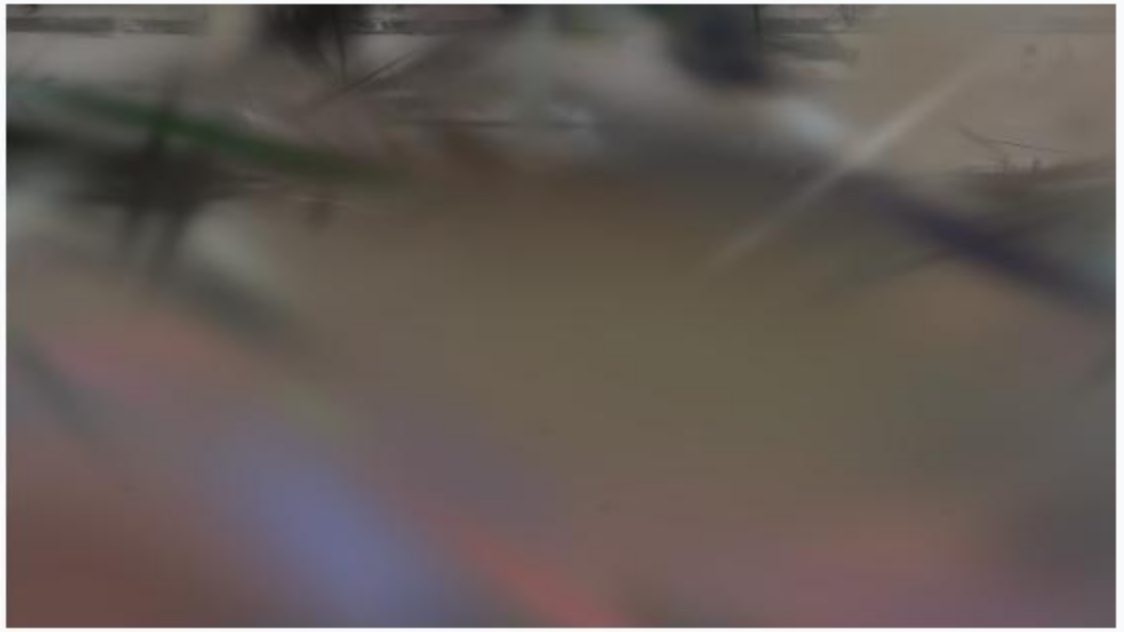
Maximum correction: 0.04156 scene units; rotation tangent 0.01827 rad

Every learned parameter hash is unchanged after rendering. Zero training steps and optimizer updates. The checkpoint does not embed the original filename order.

Training camera 0: frame_00001.png



Preprocessed source



Render with saved pose correction

PSNR: canonical 14.7467 dB; corrected 14.8001 dB.
Saved corrections do not recover the source scene structure.

Training camera 134: frame_00149.png



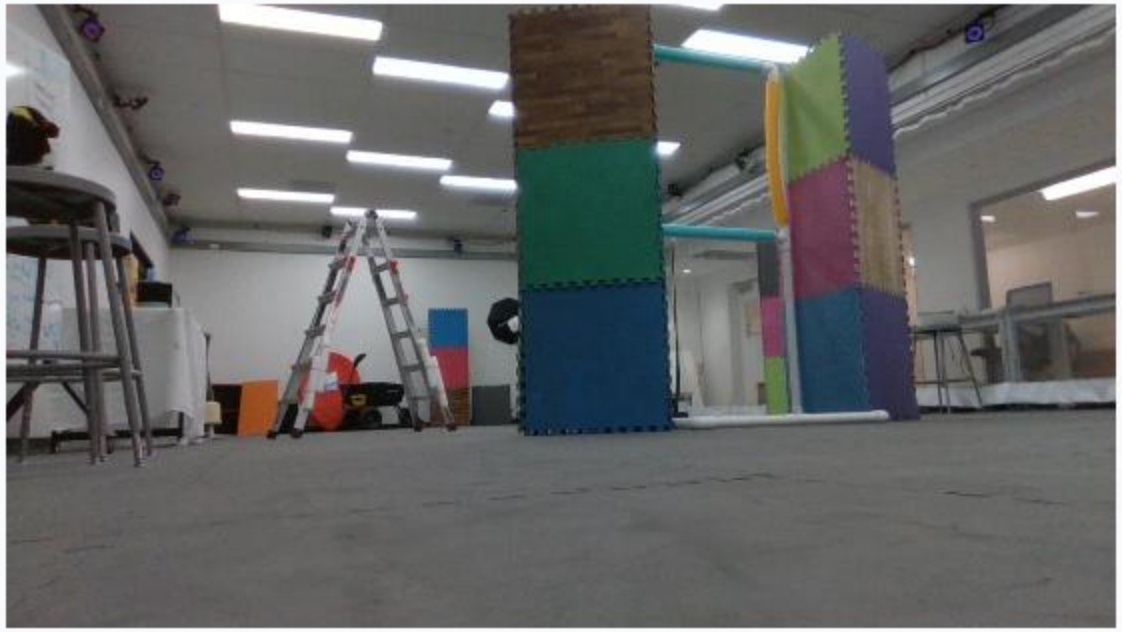
Preprocessed source



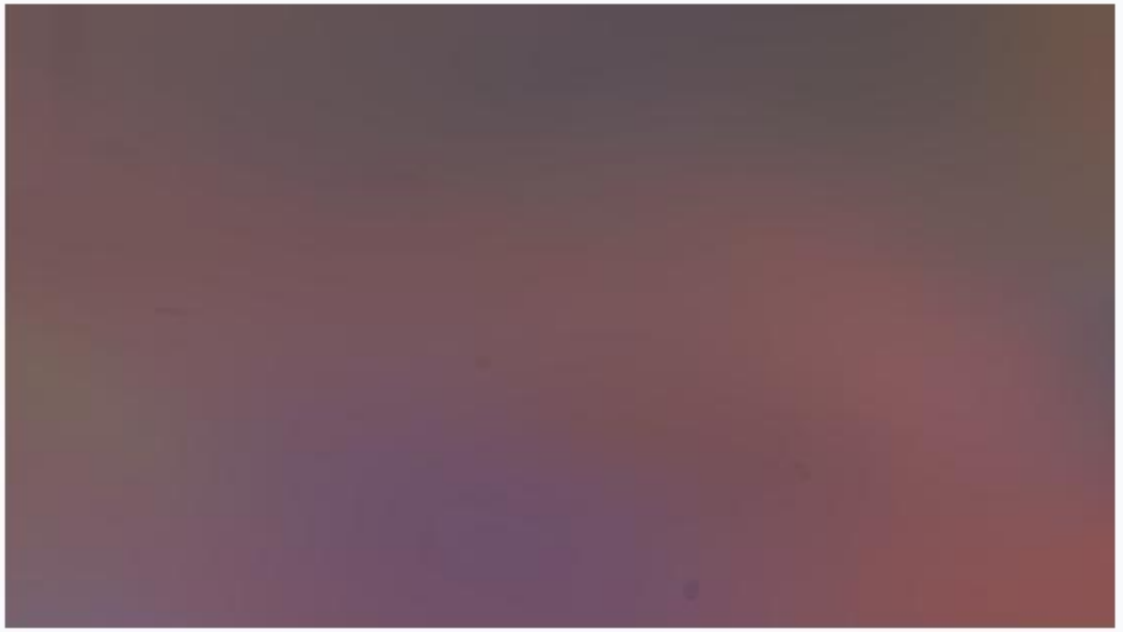
Render with saved pose correction

PSNR: canonical 13.1784 dB; corrected 13.0754 dB.
Saved corrections do not recover the source scene structure.

Training camera 269: frame_00300.png



Preprocessed source



Render with saved pose correction

PSNR: canonical 15.5847 dB; corrected 15.5810 dB.
Saved corrections do not recover the source scene structure.

Controlled Nerfstudio version comparison

1.1.3 baseline retained; 1.1.4 installed in a separate environment

Torch 2.1.2/cu118 and gsplat 1.0.0 retained exactly

Required changes only: OpenCV 4.10.0.84, viser 0.2.3, tensorly 0.9.0

Dependency check passes; all 18 RGB/depth/alpha arrays are identical

This version change did not resolve correspondence. No Gaussian restoration or optimizer-update failure was found.

Direct usefulness for Real4Real

Use case	Decision	Evidence or missing gate
Dynamics API / native integration	Usable component	B state feedback and D camera/video
Calibrated camera replay	NO-GO now	Training captures not reproduced
Original visual-policy E2E	Blocked	Compatible learned package absent
Outdoor sparse views / racing fidelity	Not established	Calibration, held-out inputs and task tests missing

Splats can supply appearance; collision geometry, calibrated physics and verified camera registration remain separate requirements.

Information needed from the authors

Matching checkpoint/config/captures bundle with hashes and reference renders

Every world transform and scale baked into the Gaussians or camera poses

Original 270-image training order, split and preprocessing steps

Exact training environment; original compatible navigation checkpoint

Until that provenance is supplied, do not use this bundle as a calibrated R4R baseline. No outdoor fidelity or real-transfer claim follows from these diagnostics.